Project Design Phase

Solution Architecture

Date	31/01/2026
Team ID	LTVIP2026TMIDS36467
Project Name	Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization
Maximum Marks	2 marks

Solution Architecture:

1️⃣ **Data Sources Layer**

- Point of Sale (POS) Systems – Transaction-level sales data (product, quantity, revenue, date, store).
- Inventory Database – Stock levels, product categories, shelf allocation.
- Store Layout Data – Product placement zones (entrance, aisle-end, eye-level shelves, checkout counters).
- Promotional Data – Discounts, campaigns, in-store offers.

2️⃣ **Data Integration & Preparation Layer**

- Data Extraction – Connect Tableau to SQL databases, Excel files, or cloud storage.
- Data Cleaning – Remove duplicates, standardize product names, handle missing values.
- Data Transformation –
- Create calculated fields (Sales Growth %, Average Basket Size, Placement Impact Score).
- Merge sales data with placement zones.
- Data Aggregation – Summarize by product, category, placement zone, and time period.

3️⃣ **Data Storage Layer**

- Centralized Data Warehouse / Data Mart
- Structured tables for:
- Sales Facts
- Product Dimensions

- Store Dimensions
- Placement Dimensions

4️⃣ **Analytics & Business Logic Layer**

- KPI Calculations:
- Total Sales
- Sales by Placement Zone
- Conversion Rate
- Revenue per Square Foot
- Before vs After Placement Comparison
- Trend Analysis (Weekly / Monthly / Seasonal)
- Correlation analysis between placement position and sales performance

5️⃣ **Visualization Layer (Using Tableau)**

Interactive dashboards built in Tableau:

- 📊 Sales by Placement Heatmap
- 📈 Trend Analysis Line Charts
- 🛒 Product Category Performance Dashboard
- 🏪 Store-wise Placement Impact Comparison
- 📍 Shelf Position vs Revenue Bar Charts

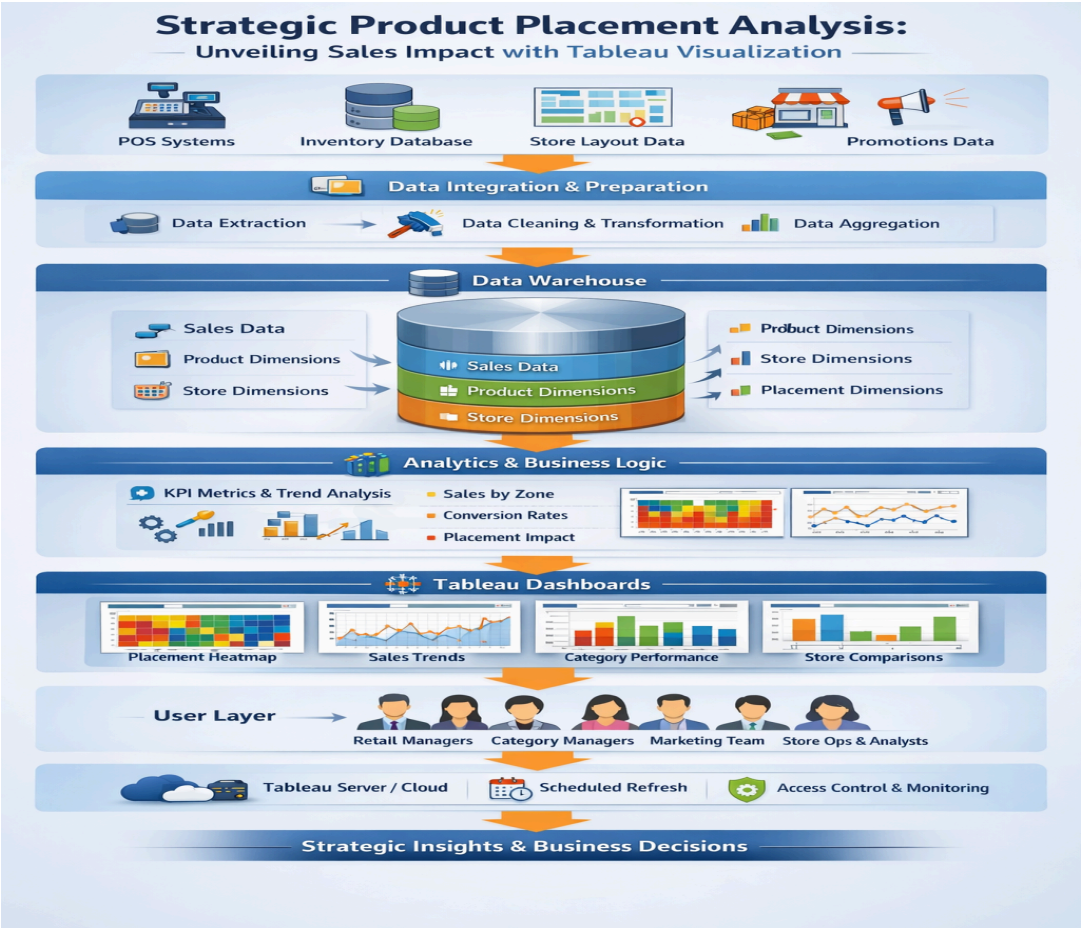
6️⃣ User Layer

- Retail Managers
- Category Managers
- Marketing Team
- Store Operations Team
- Business Analysts

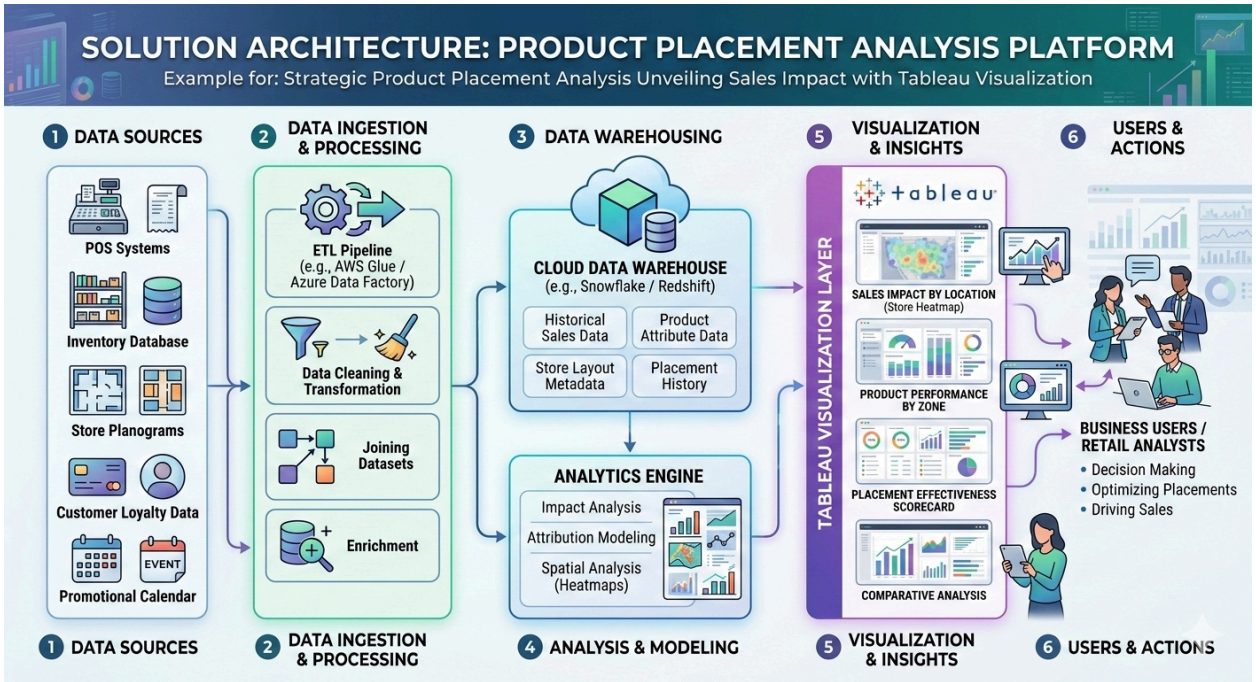
7️⃣ Deployment & Maintenance

- Publish dashboards to Tableau Server / Tableau Cloud
- Schedule automatic data refresh
- Role-based access control
- Performance monitoring and periodic optimization

🔷 Architecture Flow Summary



Example - Solution Architecture Diagram:


SOLUTION ARCHITECTURE: PRODUCT PLACEMENT ANALYSIS PLATFORM
Example for: Strategic Product Placement Analysis Unveiling Sales Impact with Tableau Visualization
1 DATA SOURCES
2 DATA INGESTION & PROCESSING
3 DATA WAREHOUSING
5 VISUALIZATION & INSIGHTS
6 USERS & ACTIONS
POS Systems
Inventory Database
Store Planograms
Customer Loyalty Data
EVENT
Promotional Calendar
ETL Pipeline (e.g., AWS Glue / Azure Data Factory)
Data Cleaning & Transformation
Joining Datasets
Enrichment
CLOUD DATA WAREHOUSE (e.g., Snowflake / Redshift)
Historical Sales Data
Product Attribute Data
Store Layout Metadata
Placement History
ANALYTICS ENGINE
Impact Analysis
Attribution Modeling
Spatial Analysis (Heatmaps)
TABLEAU VISUALIZATION LAYER
tableau
SALES IMPACT BY LOCATION (Store Heatmap)
PRODUCT PERFORMANCE BY ZONE
PLACEMENT EFFECTIVENESS SCORECARD
COMPARATIVE ANALYSIS
BUSINESS USERS / RETAIL ANALYSTS
• Decision Making
• Optimizing Placements
• Driving Sales
1 DATA SOURCES
2 DATA INGESTION & PROCESSING
4 ANALYSIS & MODELING
5 VISUALIZATION & INSIGHTS
6 USERS & ACTIONS